CPE-2301- Postlab 8

Learning Objective

- To understand memory organization and byte addressing by designing, implementing, and verifying memory read/write operations using SystemVerilog on the Basys3 FPGA board.

Instructions:

i. For your lab assignment submission on Canvas, for each group, one member should upload a single PDF to Canvas containing the complete lab work, including answers to all questions.

ii. In your solution, ensure that you clearly indicate the corresponding question number for each answer.

iii. Provide a demonstration in class for the below mentioned design to show that your lab design works 100% on a BASYS3 board. 100% working for a design means your design performs the truth table of the circuit.

Questions/Problems

1. Design and implement a 4-word × 32-bit memory module using SystemVerilog on the Basys3 board. Each 32-bit word should contain four individually addressable bytes (8-bit), providing a total of 16 byte addresses (0-15). Develop a testbench in Vivado to verify the read/write operations at different memory addresses and demonstrate the complete functionality on the Basys3 board.

[Hint: Use SW0-SW7 to enter 8-bit data and SW8-SW11 to select a memory address. Implement memory write and read operations using push buttons, BTNU (write) and BTND (read), respectively. Use LD15 and LD14 as write and read operation indicators and display the selected memory address and retrieved data in hexadecimal format using the four-digit seven-segment display. Ensure that writing to a specific byte does not modify other bytes within the same word.]

Lab Deliverables: Please submit screenshots of the complete design code, complete testbench code, Elaborated Design, simulation waveform (all test cases should be clearly visible), and constraint file (including screenshots showing any modifications made to run the corresponding lab experiment). ***An in-class demonstration of this design on the BASYS3 board is required.***